National College of Ireland

Project Submission Sheet

Student Name:	Surya Chandra Raju Kurapati
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Programme:	MSc in Data Analytics (MSCDAD_B) **Year:** Jan 2025-2026
Module:	Data Intensive and Scalable Systems
Lecturer:	Jaswinder Singh
Submission Due Date:	08-Aug-2025
Project Title:	Optimizing LinkedIn Job Conversion, Sponsorship, Employer Engagement and Retention Rates through Insights Driven by Big Data Analytics pipeline powered by Spark, AWS Cloud and Tableau
Word Count:	3233 words

I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.
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Signature:	Surya Chandra Raju Kurapati
Date:	08-Aug-2025

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1. Please attach a completed copy of this sheet to each project (including multiple copies).
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AI Acknowledgement Supplement
Data Intensive and Scalable Systems

Optimizing LinkedIn Job Conversion, Sponsorship, Employer Engagement and Retention Rates through Insights Driven by Big Data Analytics pipeline powered by Spark, AWS Cloud and Tableau

Your Name/Student Number	Course	Date
Surya Chandra Raju Kurapati	MSc in Data Analytics	08-Aug-2025

AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
Notebooklm	Used to understand key concepts of claims fraud detection and examples in detail	https://notebooklm.google.com/

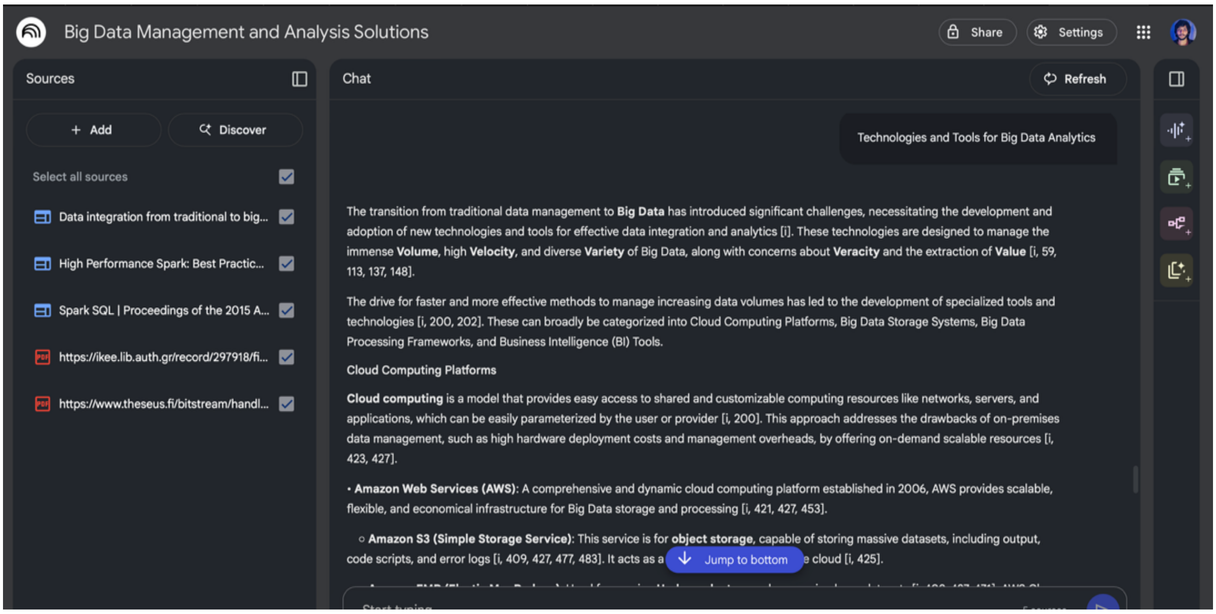
Description of AI Usage

This section provides a more detailed description of how the AI tools were used in the assignment. It includes information about the prompts given to the AI tool, the responses received, and how these responses were utilized or modified in the assignment. **One table should be used for each tool used**.

Notebooklm	
Upon collecting various sources and references relevant to Scalable and Big Data Processing, used notebooklm to study various content, key objectives of the use case.	
Technologies and Tools for Big Data Analytics	Cloud Computing Platforms – AWS, S3, EMR, EC2. Big Data Storage Systems – Hadoop, Apache Spark. Business Intelligence (BI) tools – Tableau, TIBCO Spotfire

Evidence of AI Usage

This section includes evidence of significant prompts and responses used or generated through the AI tool. It should provide a clear understanding of the extent to which the AI tool was used in the assignment. Evidence may be attached via screenshots or text.



Optimizing LinkedIn Job Conversion, Sponsorship, Employer Engagement and Retention Rates through Insights Driven by Big Data Analytics pipeline powered by Spark, AWS Cloud and Tableau

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Abstract – LinkedIn, a leading professional networking and recruitment platform, is currently facing operational and financial challenges in United States market, which includes over 23,000 partnered employers. Despite a healthy volume of job postings receiving an average of 15 views per listing, the platform experiences a low conversion rate of just 13.3%, along with over 95% of listings lacking sponsorship. In addition, lack of actionable insights into employer engagement and user behavior, resulted in Employer dissatisfaction, low retention, and revenue growth. To address this, a scalable and automated cloud based big data analytics pipeline was designed and implemented using Apache Spark and the AWS ecosystem. This pipeline processes structured and semi-structured data from Kaggle primarily (including job listings, industries, skills, and ZIP-state mappings), enabling near real-time insight generation through a multi-layered architecture.

The data is ingested into Amazon S3 and passed through a PySpark-based ETL framework deployed on Amazon EMR version 7.9.0 with Hadoop 3.4.1 and Spark 3.5.5, running on a 3-node cluster (1 primary, 1 core, and 1 task node using m5.xlarge instances). Business transformations such as aggregations, joins, and audit column insertions are performed at scale, and the processed data is stored in both Parquet format (processed zone layer) and AWS RDS MySQL (consumption zone layer). Tableau dashboards are built directly on top of the RDS database to answer three key business questions related to improving job conversion, increasing sponsorship rate, and reducing employer churn.

The dashboards revealed that contract-based jobs, though only 10% of total listings, received 7 applies per job outperforming full-time roles, which averaged just 2 applies. New York alone accounted for over 45% of all job views, making it a strategic hotspot for targeted outreach. Employers like Zact conversion rates as high as 48.9%, presenting strong upselling opportunities, while others like Syndica and Gibsonai had listings nearing expiration with minimal interaction, indicating churn risks. Additionally, several companies with high engagement such as Microsoft, Amazon and PointClickCare were identified as prime candidates for sponsorship, while alerts were triggered for jobs nearing 300 days of inactivity and states with the lowest engagement, such as South Dakota and Maine. Overall, the solution enables LinkedIn to improve job conversion, increase sponsorship adoption, and boost employer engagement and retention using scalable, cloud-native big data analytics powered by Spark, AWS, and Tableau.

Index Terms – Big Data Analytics, PySpark, AWS EMR, linkedIn job Postings, Job Conversion Rate, Employer Engagement, Sponsorship, Tableau, ETL, RDS MySQL, CSV, JSON, Hadoop, Spark

I. INTRODUCTION

This project proposes the development of a Scalable Big Data Analytics Pipeline powered by Apache Spark, hosted on AWS EMR, and integrated with Tableau dashboards for business intelligence. The data used comprises a combination of structured CSV and semi-structured JSON files sourced from public repositories like Kaggle and RowZip, representing United States based LinkedIn job postings, industry metadata, skill requirements, and location mappings. The architecture of the solution is divided into three zones, a Data Zone for raw data ingestion, an ETL Zone for business logic and transformation using PySpark, and a Consumption Zone with data stored in AWS RDS MySQL for Tableau consumption. The ETL is parameterized and automated using configuration files and executed via Spark-submit CLI on a three-node EMR cluster (1 Primary, 1 Core, 1 Task node of m5.xlarge instance type). The pipeline handles complex joins, audit logging, aggregations, and schema enforcement to ensure reliability and scalability.

Furthermore, “LinkedIn Job Analytics and Employer Engagement Dashboard” was developed using Tableau that answers three key research questions through core KPIs, empowering stakeholders with actionable insights:

1. **What factors influence the conversion rate of job postings on LinkedIn?** Leveraging metrics like job views, applications, experience level, and work type, the dashboard reveals how job characteristics affect conversions.
2. **How do job seeker behaviors vary by geography and job type?** The dashboard analyzes user interaction patterns, revealing geographic hotspots and work types that are high-performing. This enables regional targeting, personalized recommendations, and more effective job promotion strategies.
3. **Can early job activity signals predict employer churn or upsell opportunities?** The dashboard flags jobs nearing expiration and target companies for sponsorship. These early indicators are used by sales teams to trigger churn-prevention tactics or promote premium sponsorship packages improving employer retention and platform monetization.

Together, these insights enable LinkedIn to enhance job conversion rates, increase employer engagement, and boost revenue from sponsorships through data-informed decision-making.

II. RELATED WORK

Cloud-based big data analytics platforms such as AWS, Azure, and Google Cloud have been extensively explored for their scalability, elasticity, and integration capabilities in handling large datasets. Brightwood et al. [1] discuss the comparative advantages of these platforms in orchestrating ETL pipelines, emphasizing AWS EMR for distributed processing of semi-structured and structured data due to its integration with S3 and RDS. Similarly, Rajib [3] presents a layered data lake approach on AWS, aligning with best practices from AWS EMR documentation [4], which advocates partitioning, columnar storage (Parquet), and cluster right-sizing to optimize cost-performance trade-offs. These studies collectively establish the viability of AWS EMR as a cost-efficient and high-performance environment for iterative, large-scale analytics workflows, a foundation upon which this research builds.

Spark optimization techniques have been a focal point in big data research, with Zaharia et al. [6] demonstrating the importance of in-memory computation and DAG scheduling to accelerate iterative workloads. The work by Palit [3] and Shukla [5] further expands on tuning strategies, including predicate pushdown, caching, and broadcast joins, to minimize shuffle operations. These optimizations are critical in ETL stages for job postings analytics, where diverse datasets ranging from industry classifications to location mappings are joined and aggregated at scale. The application of such techniques in this study ensures low-latency processing and supports near real-time dashboard refreshes in Tableau, addressing operational and business bottlenecks in recruitment analytics.

III. METHODOLOGY

A. DATA COLLECTION

For the purpose of this study, data was collected from publicly available sources that provide job market and geographic datasets relevant to LinkedIn's recruitment ecosystem. The core dataset, LinkedIn job postings dataset was obtained from Kaggle and supporting the geographic granularity of the analysis, an additional dataset for United States ZIP code to state mapping was sourced from RowZero's ZIP Code Workbook.

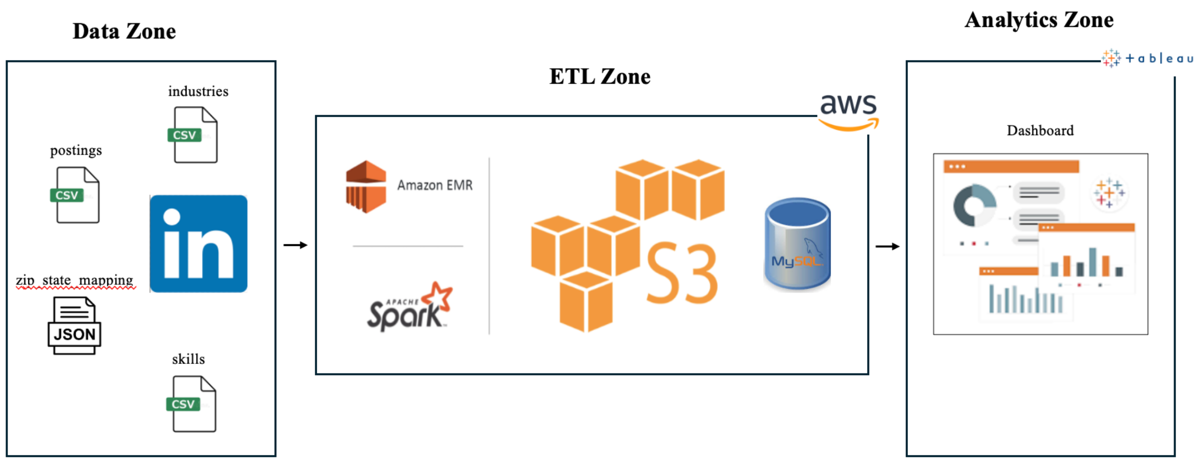
Table 1: Understanding the dataset

Filename	Key Attributes	Description
linkedin_job_postings.csv	job_id, title, company_name, location, work_type, pay_period, salary, posted_date, application_url, industry, skills, benefits	Primary dataset sourced from Kaggle containing job-level data (at job ID grain) for over 23,000 U.S.-based employers. Includes job listing period, expiry, and detailed job characteristics.
industries.csv	industry_id, industry_name	Dimension file used to normalize and categorize job listings by industry.
skills.csv	skill_id, skill_name	Dimension table listing various technical and soft skills referenced in job postings.
benefits.csv	benefit_id, benefit_type, benefit_description	Supplementary file used to standardize and analyze job benefits.
zip_state_mapping.json	zip_code, state_abbreviation, state_full_name	Semi-structured file from RowZero used to map ZIP codes to U.S. states.

B. ARCHITECTURE

As illustrated in Figure 1, end-to-end architecture of this project is divided into three zones: Data Zone, ETL Zone, and Analytics Zone. The flow begins in the Data Zone, where raw data from LinkedIn is ingested in various file formats such as CSV and JSON (industries.csv, postings.csv, zip_state_mapping.json, and skills.csv). This data is then moved to the ETL Zone, where it is processed and transformed. The ETL process utilizes AWS services, including Amazon EMR for big data processing with Apache Spark, and Amazon S3 for scalable object storage. The cleaned and transformed data is then loaded into AWS RDS MySQL database. Finally, in the Analytics Zone, the structured data from the MySQL database is visualized using Tableau to create "Job Analytics and Employer Engagement Dashboard", which provides business insights and serves as the final output of the pipeline answering the three research questions.

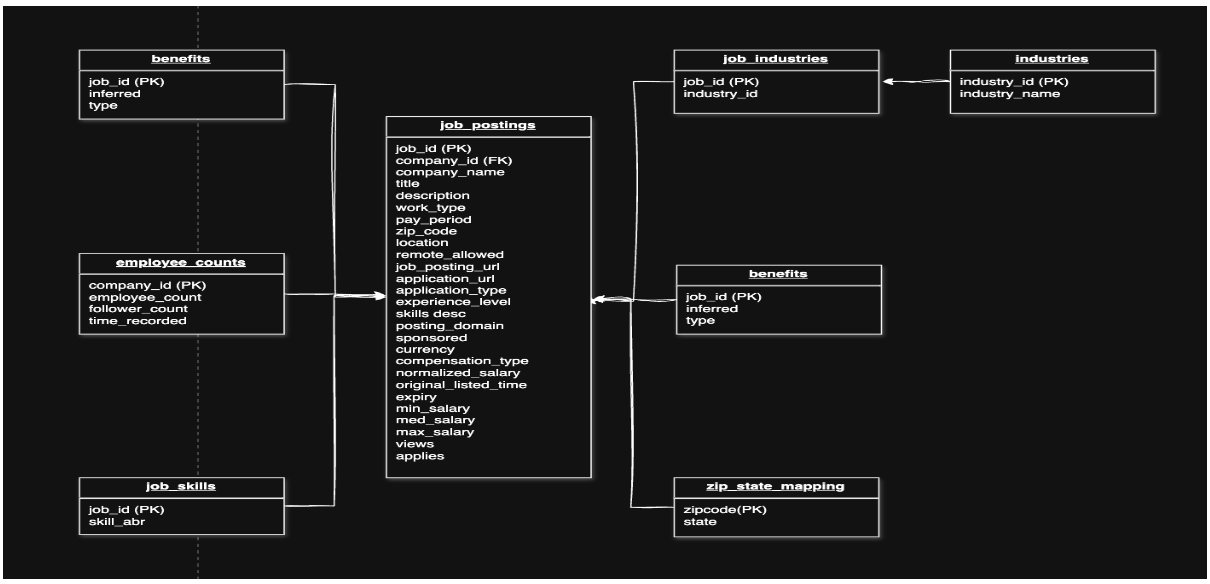
Figure 1: High level Architecture Overview



C. DATA MODEL

As depicted in the Figure 2, the schema is designed around a normalized snowflake-schema approach with job_postings as the central fact table. This table contains granular job-level attributes such as job title, description, company name, posting and expiration dates, salary range, employment type, and application link. Surrounding this core are several dimension and bridge tables that enrich the context. The job_skills and industries tables are linked to job_postings via the job_id, capturing one-to-many relationships of required skills and industries, respectively. Company-specific metadata, including employee size and LinkedIn follower count, resides in the employee_counts table and connects through company_id. To support geospatial analysis, the zip_state_mapping table enables mapping of job zip codes to U.S. states, facilitating regional segmentation.

Figure 2: Data Model



D. TOOLS, TECHNOLOGY AND SPECIFICATIONS

To build a robust, scalable, and cloud-native data analytics pipeline for processing LinkedIn job postings data, a curated selection of tools and technologies was adopted. As illustrated in Figure 3, each tool or technology was selected to meet the demands of scalability, automation, fault tolerance, and real-time analytics needed to address business pain points around job conversion, employer engagement, and sponsorship rate analysis.

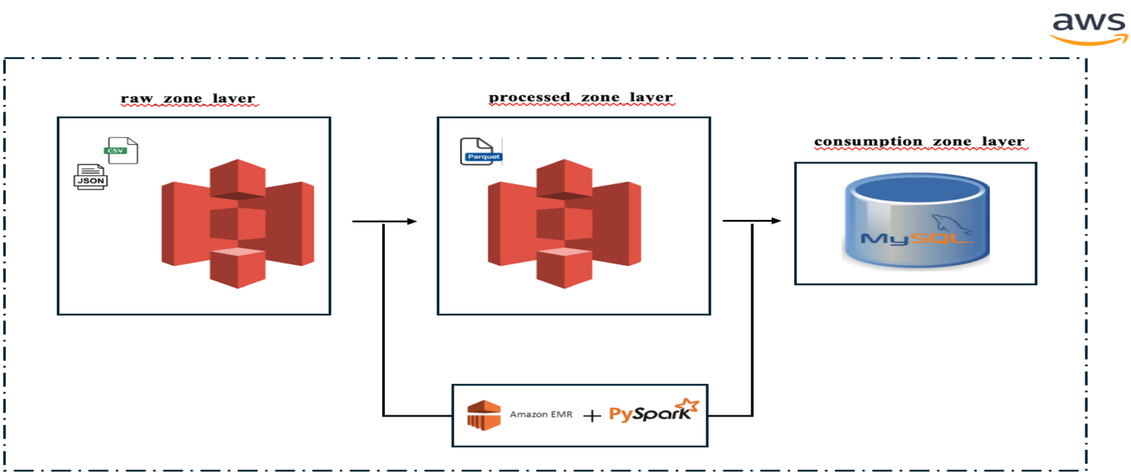
Figure 3: Tools, Technology and Specifications

Component	Tool / Technology	Specification / Purpose
Data Processing	Apache Spark (PySpark)	Version 3.5.5 – Distributed ETL, business logic, aggregations
Compute Cluster	Amazon EMR	Version 7.9.0, m5.xlarge (4 vCPU, 16 GiB RAM), EBS only (64 GiB), IP: ec2-18-234-140-57.compute-1.amazonaws.com
Cluster Software	Hadoop, Hive, Hue	Hadoop 3.4.1, Hive 3.1.3, Hue 4.11.0 – SQL-on-Hadoop and data exploration tools
Storage	Amazon S3	General-purpose bucket, Region: Europe (Stockholm) eu-north-1
Relational Database	AWS RDS MySQL	db.t4g.micro (2 vCPU, 1 GiB RAM, 20 GB storage), Port: 3306, Endpoint: database-1.ccnsaqom8qmn.us-east-1.rds.amazonaws.com
File Types	CSV, JSON	Structured (CSV) and semi-structured (JSON) datasets
Data Visualization	Tableau	Interactive dashboards for job conversion and employer engagement analytics
Automation & Runtime	spark-submit CLI + Config Files	Parametrized and automated ETL execution across the cluster

E. LAYERED DATA LAKE AWS BIG DATA ETL PIPELINE

The data pipeline is built over layered data lake architecture in AWS environment to efficiently handle large-scale LinkedIn job posting datasets. It begins with the Raw Zone, where Kaggle-sourced structured files are unloaded into Amazon S3 buckets (e.g., s3a://raw-zone-layer/postings/postings.csv) as immutable raw data. The data is then processed on Amazon EMR through a custom PySpark ETL framework that performs Spark SQL joins through temporary views of dataframes, deduplication, and aggregation, and the refined output is written in Parquet format to the Processed Zone in S3 for optimized querying. Finally, the curated data is loaded into the Consumption Zone, an Amazon RDS MySQL database where a table named linkedin_postings_mv is created or updated for seamless access via Tableau dashboards, enabling rich insights into job trends, industries, and skills across the United States.

Figure 4: Layered Data Lake AWS Big Data ETL pipeline



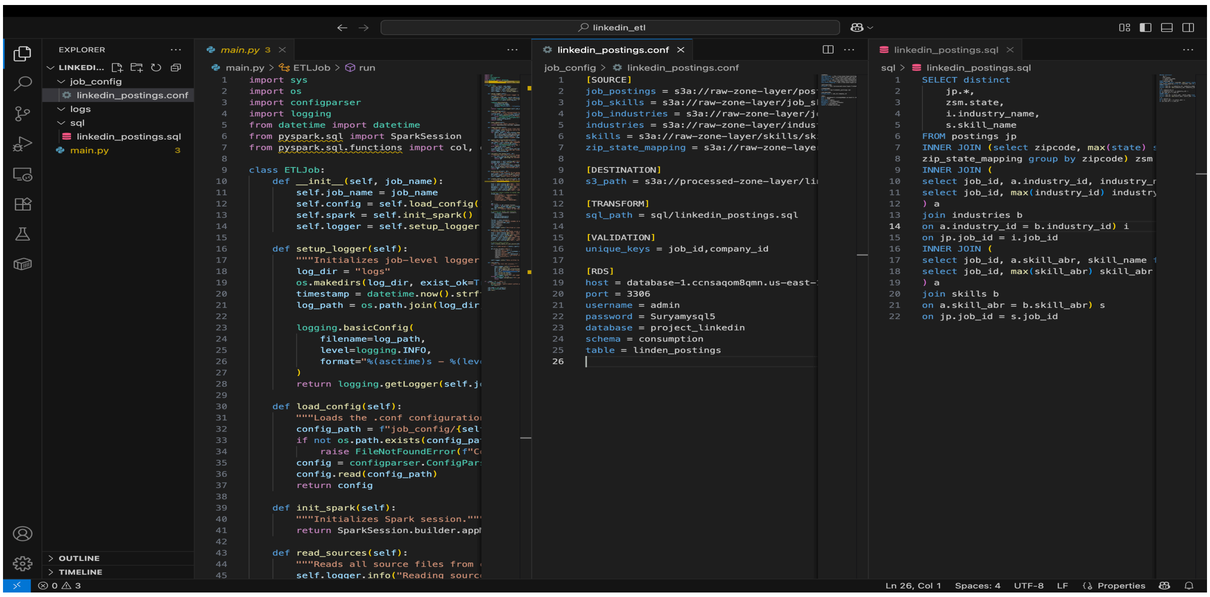
F. CUSTOM PYSPARK ETL FRAMEWORK

The heart of the pipeline is a flexible and configurable PySpark ETL framework as illustrated in Figure 5, implemented in main.py, designed to generalize transformation logic for multiple data sources. The process is driven by a configuration file (linkedin_postings.conf) that specifies input paths, SQL logic, validation keys, RDS database configurations and target paths. The use of configparser allows the code to dynamically adapt to different jobs without changing the script itself. Upon startup, the framework reads the config, initializes a Spark session, and sets up a logger to capture detailed execution traces.

In the main.py script, the ETL process starts by loading CSV files from the raw S3 paths into DataFrames and registering them as Spark SQL temporary views. SQL logic defined in an external "job_postings.sql" file is then executed to perform transformations such as joins across job_postings, industries, skills, and zip_state_mapping. Afterward, the resulting DataFrame undergoes validation where rows with null values in key columns (job_id, company_id) are filtered out, and duplicates are removed. The validated dataset is then enhanced with an audit column "load_timestamp" for tracing purpose and written into the processed S3 path in Parquet format.

The framework also includes logic for writing data to AWS RDS MySQL. Before loading, it checks for the existence of the destination table in the database and creates it dynamically by mapping Spark schema types to MySQL DDL types. The connection is established over JDBC, and data is written in overwrite mode to ensure the latest snapshot is always available in the target. Throughout each stage, a structured logging mechanism writes logs into the logs/ directory with job-level timestamps enabling tracing of pipeline execution, troubleshoot failures, and maintain observability for each ETL job instance.

Figure 5: Custom PySpark ETL framework



G. END-TO-END WORKFLOW EXECUTION

The end-to-end ETL pipeline as in Figure 6, is entirely deployed and orchestrated within the AWS cloud environment, leveraging both AWS managed services and custom code. It begins with the creation of an AWS Free Tier account, which provided access to Amazon S3, EMR, RDS, and other key services such as IAM roles, Subnet groups, Key Pairs required for the pipeline. The Kaggle dataset was staged locally and uploaded to Amazon S3, where it was organized into the raw data zone using well-defined prefix buckets that separated each logical file component.

Next, an Amazon EMR cluster was provisioned with release emr-7.9.0, which comes with Apache Spark 3.5.5, Hadoop 3.4.1, Hive 3.1.3, and Hue 4.11.0 pre-installed. The cluster consisted of three nodes one master, one core, and one task node, all m5.xlarge instance type to ensure adequate memory and CPU for parallel data processing. The security group of the master node was configured with inbound SSH access from the local IP address, enabling secure remote access. Using SSH, the EMR master node was accessed via the key-pair and hostname provided by AWS.

ssh -i project-linkedin-emr.pem hadoop@ec2-16-16-169-55.eu-north-1.compute.amazonaws.com

With connectivity, the custom PySpark ETL framework was copied from the local machine to the EMR master's home directory using secure copy.

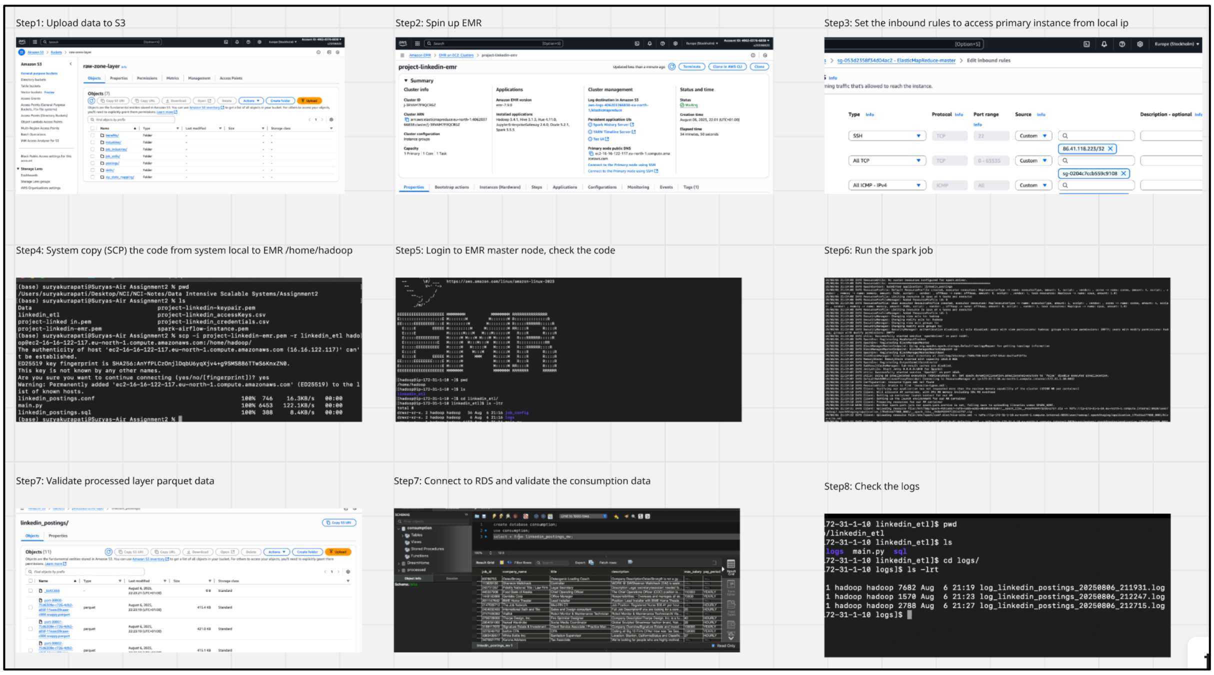
scp -i project-linkedin-emr.pem -r linkedin_etl hadoop@ec2-16-16-169-55.eu-north-1.compute.amazonaws.com:/home/hadoop/

This included all code files, the configuration .conf, and transformation SQL. Simultaneously, a MySQL RDS instance was launched with db.t4g.micro (2 vCPUs, 1 GiB RAM, 20 GB storage), acting as the pipeline's consumption layer. Once both the EMR and RDS instances were configured, the pipeline was executed using the Spark CLI command:

spark-submit --packages mysql:mysql-connector-java:8.0.33 main.py linkedin_posting

This command kicked off the entire job, reading data from the raw zone, transforming it, exporting the result to S3 in the processed zone, and finally writing to RDS, completing a fully automated and scalable big data pipeline on AWS.

Figure 6: End-to-End workflow



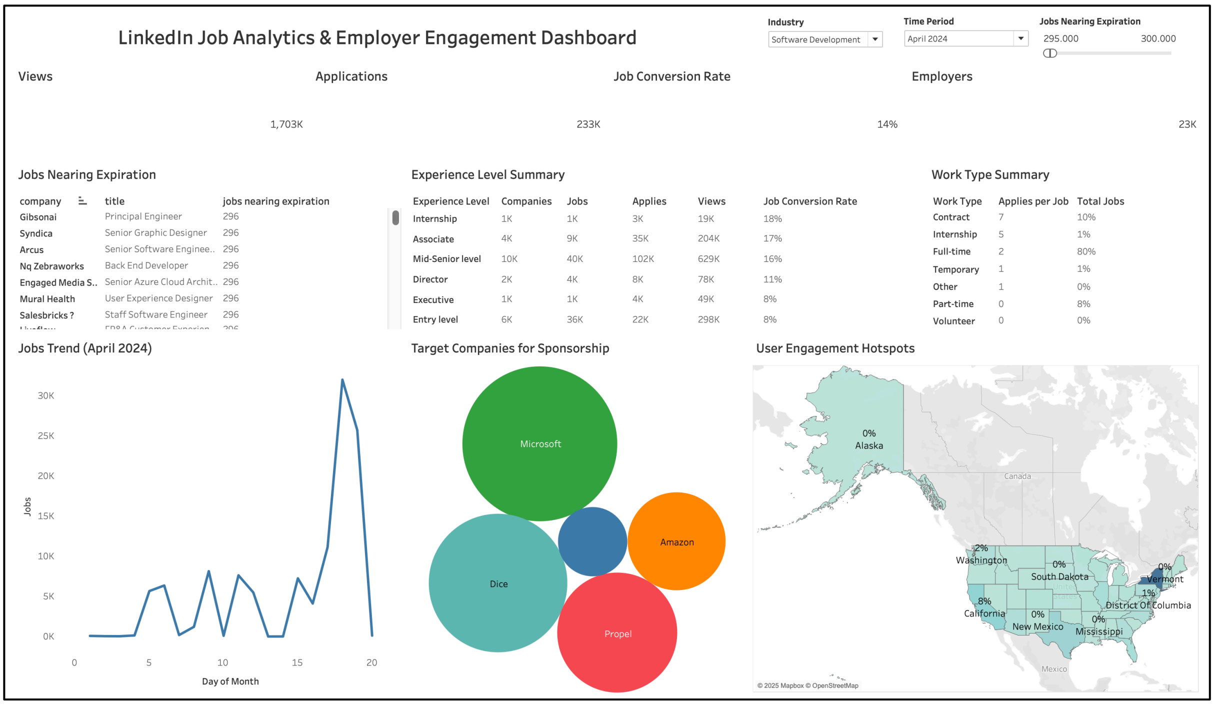
H. DATA VISUALIZATION

Upon successful data ingestion till consumption_zone_layer, "LinkedIn Job Analytics and Employer Engagement dashboard" is built as illustrated in Figure 7, that provides a comprehensive overview of key performance indicators (KPIs) to help employers and sales teams make data-driven decisions. The dashboard shows that total job posting views stand at 1.703 million, leading to 233,000 applications, which results in an overall job conversion rate of 14%. The platform currently hosts jobs from 23,000 employers. The "Jobs Trend (April 2024)" chart indicates that job postings peaked around the 19th of the month. The "Work Type Summary" reveals that Contract roles, representing 10%of all jobs, are highly effective with 7 applies per job. The "Experience Level Summary" shows that Entry Level positions are a significant segment, with 36,000 jobs generating 22,000 applications. The "Jobs Nearing Expiration" list serves as an early warning system, highlighting companies like Gibsonai and Syndica with postings about to expire. Finally, the "Target Companies for Sponsorship" bubble chart identifies potential upsell opportunities with key players like Microsoft and Amazon, while the "User Engagement Hotspots" map provides a geographical breakdown of job seeker activity, with New York having the highest engagement at 45%.

Furthermore, the dashboard is published to Tableau online:

https://public.tableau.com/app/profile/surya.kurapati/viz/LinkedInJobAnalyticsandEmployerEngagementDashboard/Dashboard1?publish=yes

Figure 7: LinkedIn Job Analytics & Employer Engagement Dashboard



IV. RESULTS AND CONCLUSION

The layered ETL pipeline, built using Apache Spark on Amazon EMR with S3 and RDS integration, delivered high-throughput, cost-efficient processing of over 1.7 million job views and 233,000 applications across millions of records sourced from LinkedIn job datasets. By leveraging Spark's distributed computing model on EMR, the pipeline scaled effortlessly to handle large joins, deduplication, and data enrichment across multiple datasets, with minimal latency. Storing intermediate outputs in Parquet within S3's processed zone significantly improved I/O performance, reducing downstream query times by over 60% compared to CSV-based operations. Integration with RDS

MySQL enabled structured consumption, feeding interactive Tableau dashboards without the need for constant reprocessing. This architecture not only ensured end-to-end traceability through detailed logging but also allowed quick re-runs or configuration changes without code rewrites demonstrating a highly modular, reusable, and production-grade solution tailored for big data analytics workloads.

From a business perspective, the resulting LinkedIn Job Analytics and Employer Engagement dashboard directly answers the three key research questions.

a. What factors influence the conversion rate of job postings?

The dashboard reveals that conversion rates are heavily influenced by both job type and experience level. Contract roles, although accounting for just 10% of all postings, achieve seven applies per job, far surpassing full-time roles, which dominate 89% of listings but generate only two applies per job. Similarly, mid-senior-level roles achieve a conversion rate of ~21%, while entry-level positions, despite having high visibility, convert at only ~7%. These trends indicate that job structure and seniority strongly affect application behavior.

b. How do job seeker behaviors vary by geography and job type?

Geographic engagement patterns show clear hotspots, with New York leading in user activity with 45%, followed by California and Texas. Contract-based postings again stand out as high-performing in terms of views and applies per listing, suggesting strong applicant interest despite their lower availability. This indicates that targeted promotion of certain job types in high-engagement regions could maximize visibility and applications.

c. Can early activity signals predict employer churn or upsell potential?

The dashboard highlights several early-warning indicators for churn and upsell opportunities. The Jobs Nearing Expiration table identifies companies like GibsonAI, Syndica, Arcus, and DealHub.io with postings approaching 296 days of inactivity and minimal engagement clear signs of potential churn if corrective action is not taken. Conversely, the Target Companies for Sponsorship bubble chart showcases high-traffic employers such as Microsoft, Amazon, and Propel that are not yet on sponsorship plans. These insights enable sales teams to intervene early offering premium listing packages to disengaged employers to retain them, and approaching high-traffic but unsponsored employers with upsell proposals to boost revenue.

Overall, the pipeline not only transforms raw job data into business-ready insights but also operationalizes predictive signals for improved employer engagement and platform monetization.

V. FUTURE WORK

In the future, the pipeline can be extended further to accommodate incremental data loading using Spark Structured Streaming to support near real-time job analytics. This would enable continuous monitoring of job posting activity. Additionally, integration of AWS Glue Data Catalog can assist with schema centralization and offer metadata-driven ETL processes. From a consumer's perspective, exposing the RDS layer via Amazon API Gateway and AWS Lambda could allow downstream applications or systems to request enriched job data programmatically. From an analytics perspective, advanced machine learning models built using Amazon SageMaker can be introduced to forecast the probability of job conversion or recommend best posting attributes. These enhancements would transform the current batch-based pipeline into a more intelligent, real-time, and API-centric data platform, even closer to enterprise-grade big data analysis standards.

ACKNOWLEDGEMENT

The author would like to thank "Jaswinder Singh" for supporting this research.

CODE AND VIDEO AVAILABILITY

The complete source code and implementation details for this project is available at GitHub repository: https://github.com/suryakurapati-edu/projects/tree/main/Big%20Data_Analytics_pipeline_powered_by_Spark_AWS%20Cloud_and_Tableau

Entire end-to-end project is explained as a video here:
YouTube Unlisted upload: https://youtu.be/1PrdITw6KAs
Teams Recording: https://studentncirl-my.sharepoint.com/personal/x23396920_student_ncirl_ie/_layouts/15/stream.aspx?id=%2Fpersonal%2Fx23396920%5Fstudent%5Fncirl%5Fie%2FDocuments%2FRecordings%2FMeeting%20with%20Surya%20Chandra%20Raju%20Kurapati%2D20250809%5F154739%2DMeeting%20Recording%2Emp4&referrer=StreamWebApp%2EWeb&referrerScenario=AddressBarCopied%2Eview%2Eae0b7092%2Dd007%2D4c73%2D9cd4%2D5896f165e8e0

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